

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

Computer Vision – 21CSE454T

UNIT 2: SHAPES AND REGIONS | SESSION 7: HANDLING OCCLUSION

SLO 1: Understand and explain the challenges posed by occlusion in object detection and shape analysis.

SCENARIO ANALYSIS SOLUTIONS

Situation: In a traffic monitoring system, one car partially blocks another car.

1. What problem does this create for the object detection system?

It causes boundary fragmentation, merges two distinct object silhouettes into a single composite blob, and hides critical shape features like license plates, total dimensions, or geometry.

2. Why might the system incorrectly identify the hidden car?

Due to missing visual information and broken contours, the feature extractor may misinterpret the visible portion as noise, background, or mistake the combined vehicles for a single oversized vehicle.

SECTION 4 – IDENTIFY THE CHALLENGE SOLUTIONS

Problem	Correct Match / Description
Partial Occlusion	B (Only part of the object is visible)
Full Occlusion	A (Object completely hidden)
Detection Error	C (System misidentifies object due to missing information)

SECTION 5 – APPLICATION THINKING SOLUTIONS

Application	Example of Occlusion
Traffic monitoring	A large truck or bus blocking a smaller car from the camera view.
Face recognition	A person wearing a face mask, sunglasses, or a hat covering facial features.
Medical imaging	Bones or dense tissue overlapping and obscuring internal organs or blood vessels.

REFLECTION SOLUTION

Why is occlusion considered a major challenge in computer vision?

Occlusion disrupts continuous object boundaries and causes a critical loss of visual data, preventing standard feature extraction and shape models from accurately recognizing, separating, and classifying objects.